Combinatorics: Diagnostic C

Time allowed: 45 minutes **Total:** 14 diagnostic marks

Instructions Answer both questions. Show full working and give complete proofs. No calculator, internet, AI assistance or hints during an independent diagnostic. Use separate answer paper if required.

Candidate: _____ Date: _____

Question 1 [7 marks]

COMB-01

Write the integers $1, 2, 3, \ldots, 2026$ in a row. In one move, choose any two adjacent entries a and b and replace them with the single entry

$$a + b + ab.$$

Continue until only one entry remains.

Determine the final entry, and prove that it does not depend on the choices of adjacent pairs.

Question 2 [7 marks]

COMB-02

In a group, every pair of distinct people are either mutual friends or mutual strangers. Find the *smallest* positive integer n with the following property:

> *Every group of n people contains either three pairwise mutual friends or three pairwise mutual strangers.*

Prove that your value works, and that a smaller value does not.

End of paper. Write your answers on separate paper, labelled with the question number.